



Mechanical Spotted Lanternflies Trap for Vineyard Harvest Protection

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Introduction

Spotted lanternflies (SLFs) are an invasive vineyard pest. They cause vine damage, reduce fruit quality, and contaminate harvests. The problem is worsening, and a Cornell study estimates that losses could reach \$8.8 million in the grape industry within three years.¹

Region	Production (ton)	Value (\$/ton)	Value lost after first year (\$/ton)	Production lost in second year (ton)	Value lost after second year (\$/ton)	Production lost in subsequent years (ton)	Value lost in subsequent years (\$/ton)
Finger Lakes	7,352	12,091,847	362,755	221	949,210	588	2,079,798
Lake Erie	121,545	39,064,034	1,171,921	3,646	3,066,527	9,724	6,719,014
Total NYS	128,897	51,155,882	1,534,676	3,867	4,015,737	10,210	8,798,812

Table 1: Grape production assumptions for selected grape varieties in Finger Lakes and Lake Erie.³

Our Goal

This project focuses on developing a mechanical system that attracts SLFs away from vines and contains them before they enter the harvest process.

The Subproblem

Spotted lanternflies return to grapevines in the days right before harvests when pesticides cannot be used. Mechanical harvesters often collect insects with the fruit, and a small number of SLFs can contaminate an entire load. At this stage, **pesticides cannot be used due to food safety concerns, and manual removal is too slow for large vineyards.** This is the problem we chose to address. To do so, we designed a low-cost, customizable, mechanical trap to keep SLFs off the grapevines right before harvest. Current solutions are expensive, labor-intensive, and harmful to the environment.



Figure 1: Lanternflies on grapevines ².

Functionality

Our final prototype is composed of 3 components

- **Funnel:** An octagonal opening with 8 interlocking flaps mounted underneath.

- **Funnel (cont.):** This creates a one-way opening, the flaps open with minimal force applied by the SFL, then interlock to block anything from exiting back through. A second octagon and elastic lie underneath to provide a restoring force that interlocks the flaps.
- **Case with Drawer:** A box-shaped case featuring a drawer with slides for easy opening. Allows convenient disposal of dead SFLs and can hold Tree of Heaven sap or other SFL attractors as bait.
- **Adjustable Post:** Case mounts on to an adjustable post for mounting at various heights and locations. The post uses two rails with holes - one slides inside the other and locks at the desired height by inserting a pin through aligned holes.

Prototype

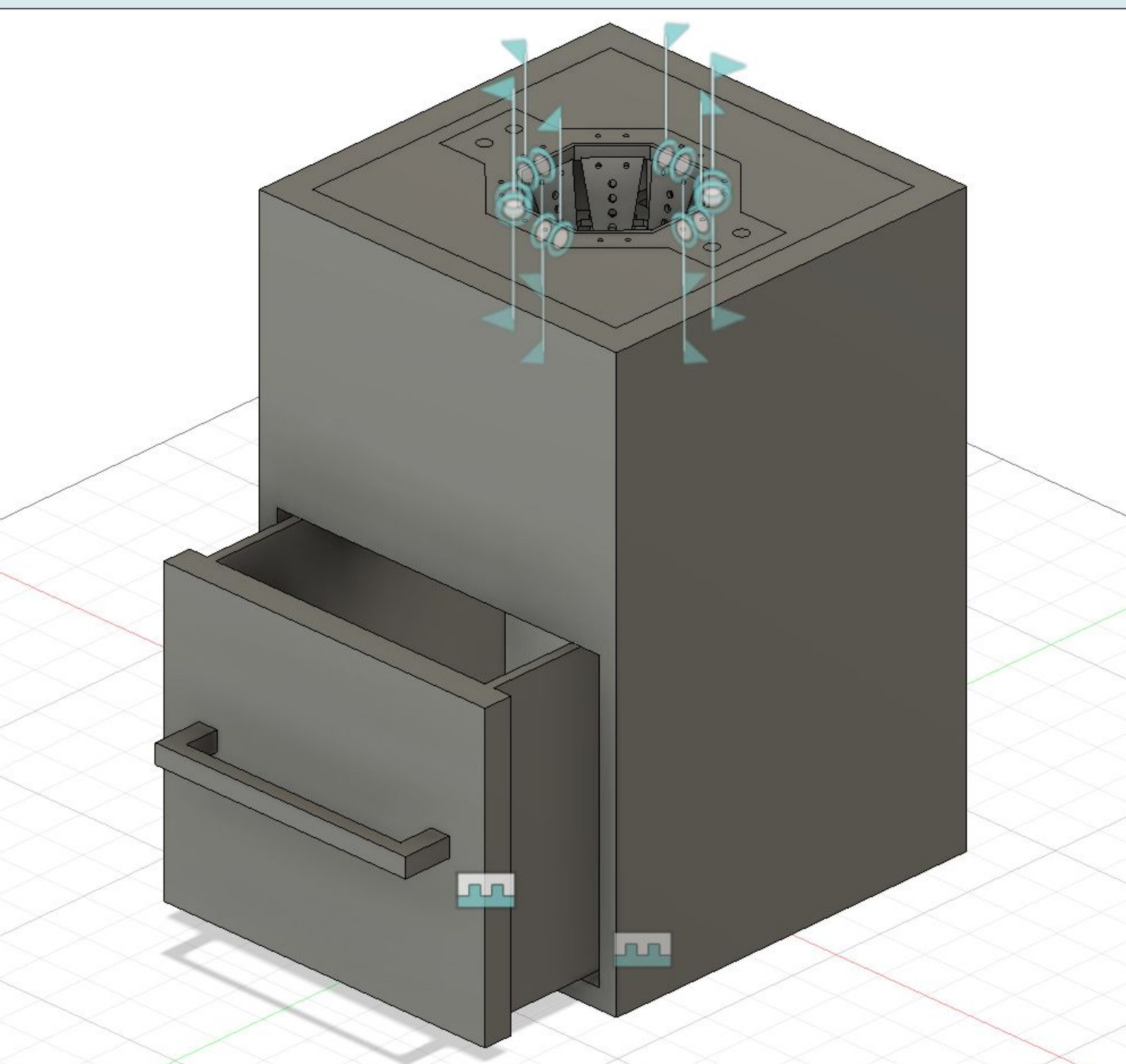


Figure 2: CAD model of the entire box.

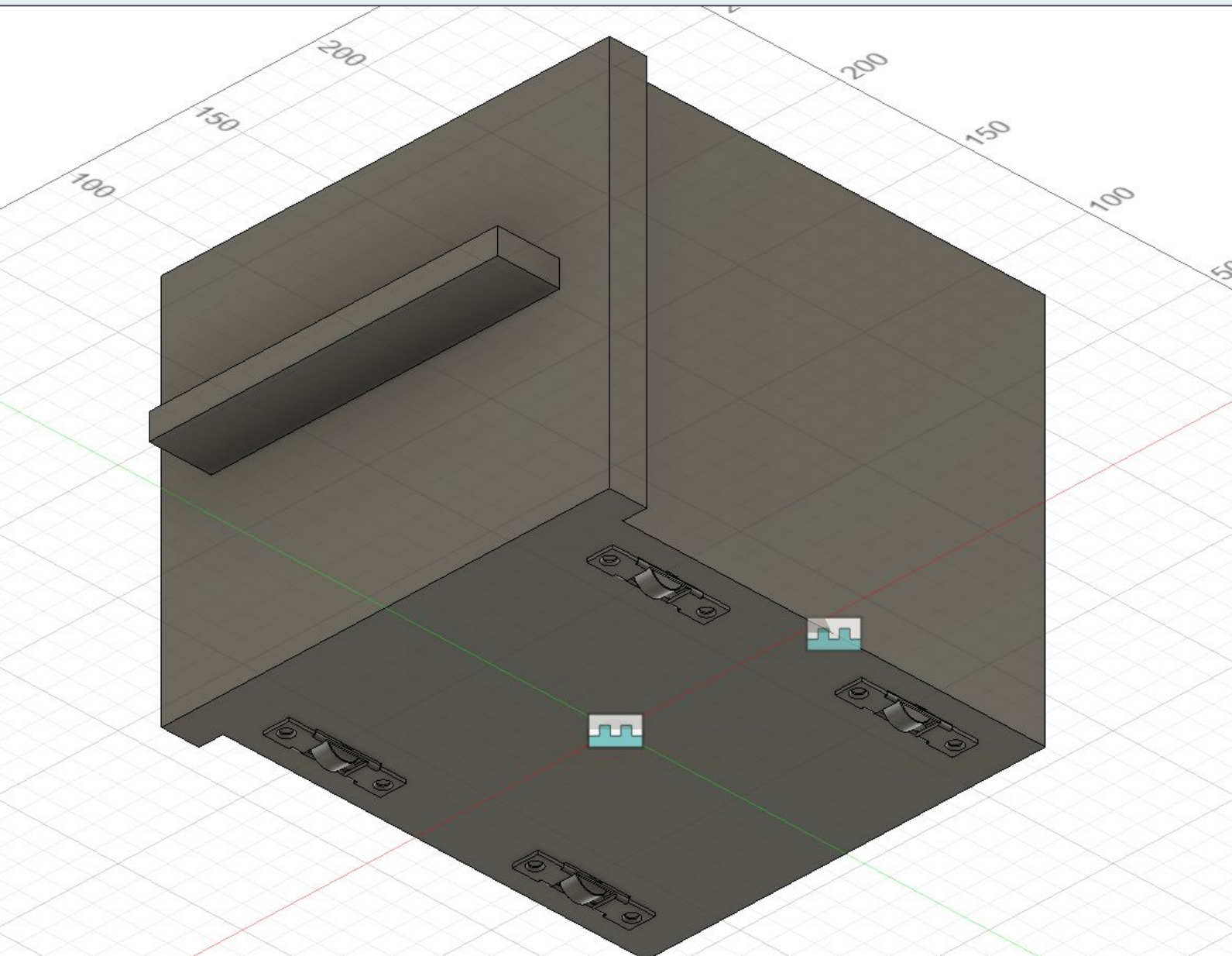


Figure 3: Bottom view of CAD model for the drawer & wheel system.

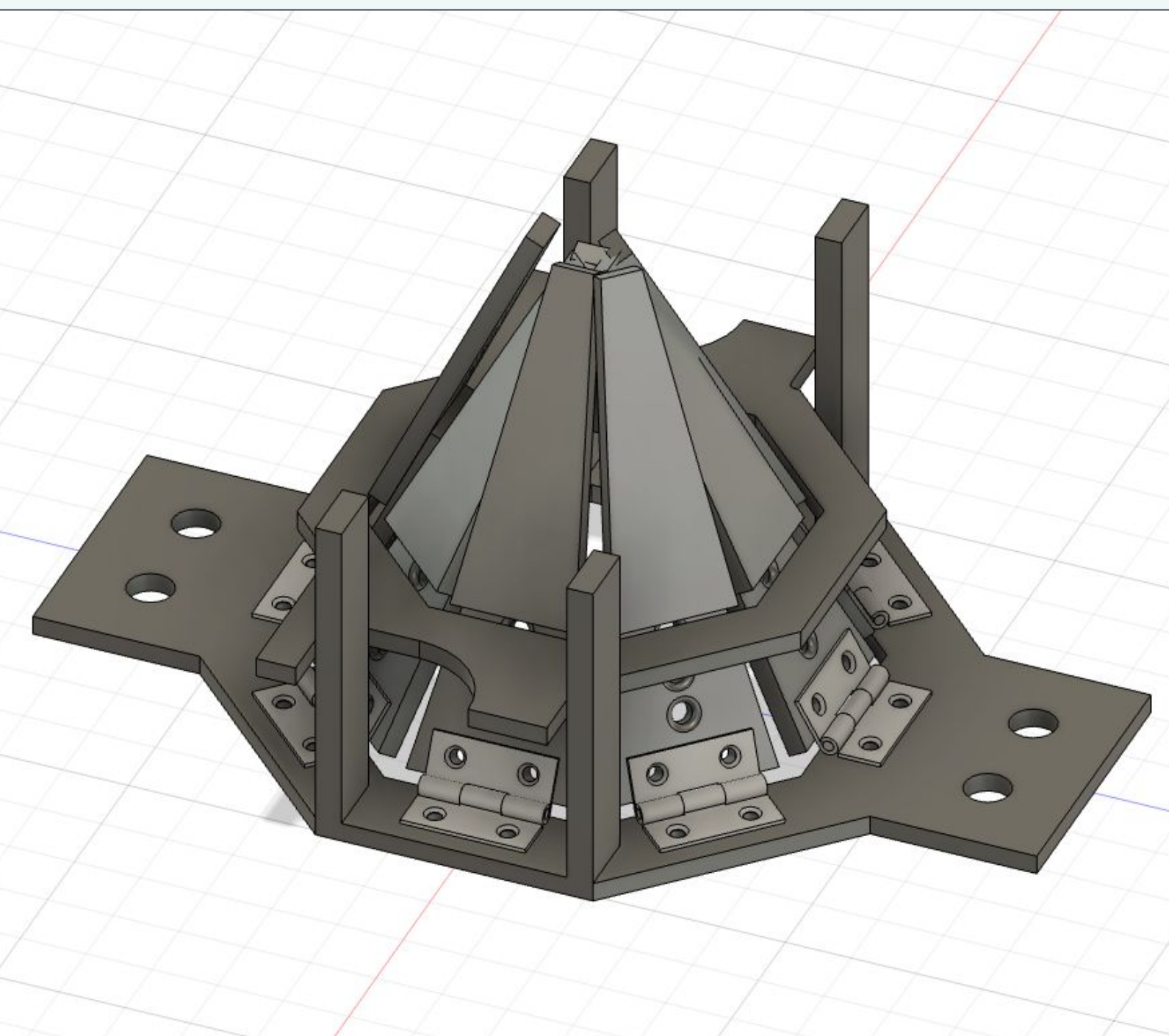


Figure 4: CAD model of underside of the funnel-flap system.



Figure 5: Close up of the funnel-claw system.

Tests & Results

- **Flap Range of Motion and Default Position:** Assessed whether the funnel opening could open with the small force provided by a SFL. A spring balance measured the force as a model fly was lowered into the opening.
- **Drawer Weight Capacity:** Assessed whether the drawer operated smoothly when loaded to its full capacity (estimated at 5.4 kg). Test weights totaling 5.4 kg were used, and repeated sliding operations were performed to verify smooth function under load.
- **Pole Height Adjustment:** Assessed whether the pole could support the trap when driven into the ground. Pole was dug into the ground at different depths to observe its stability.

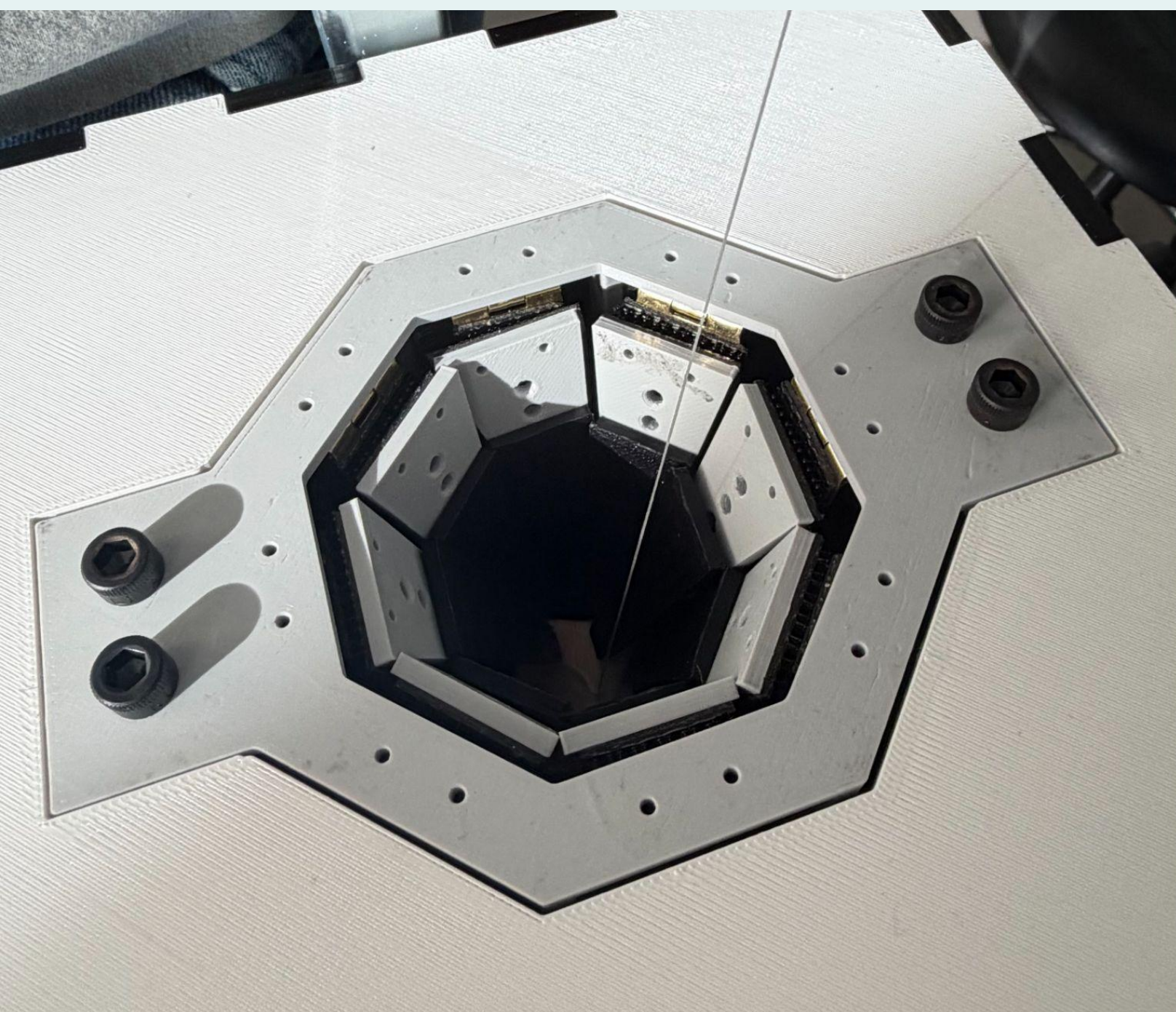


Figure 6: Top view of the funnel opening preventing the mock bug from exiting.



Figure 7: Drawer weight test.

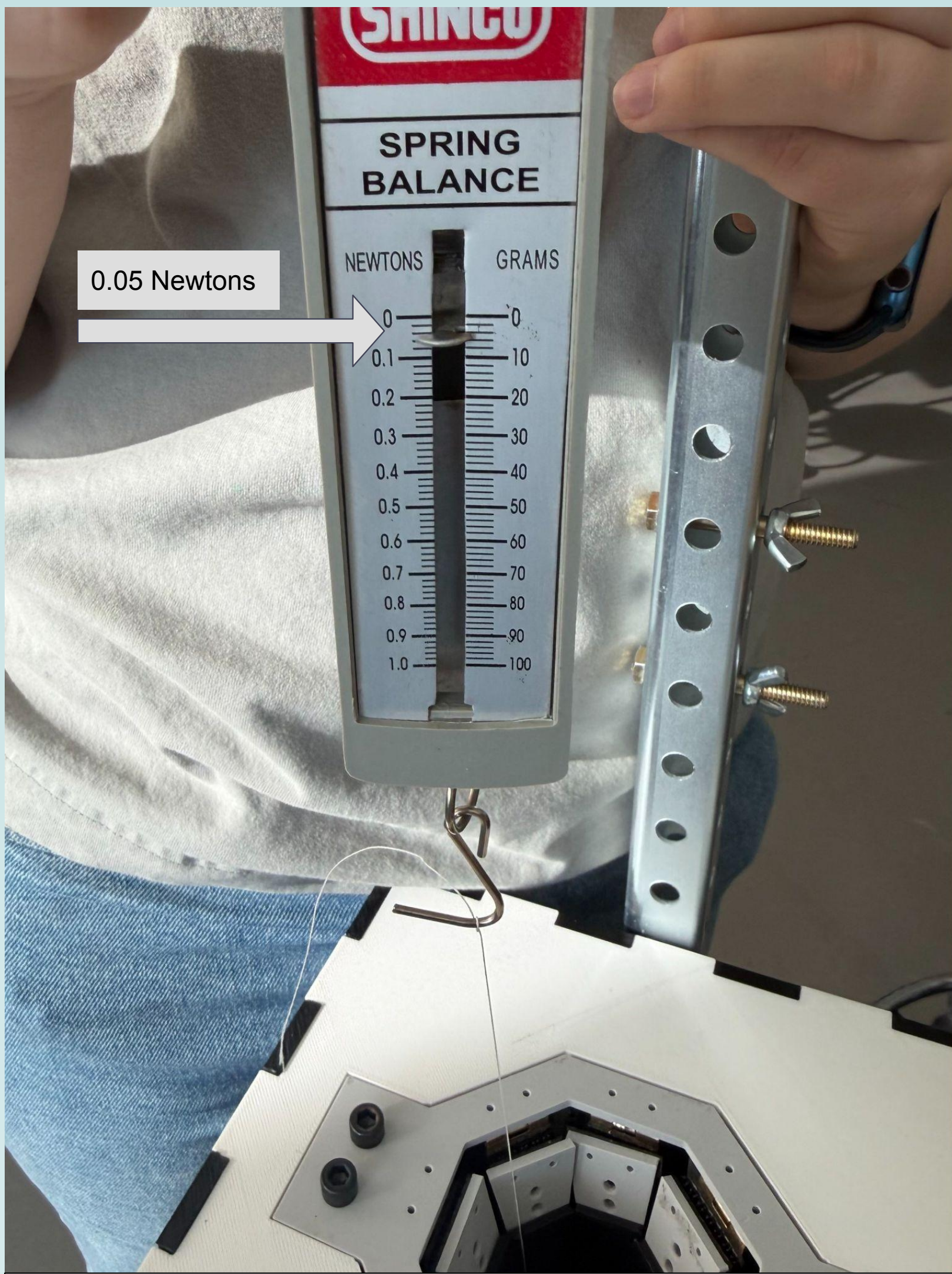


Figure 8: Close up of mock bug being used to measure the trigger force required for the funnel-claw system to open. Measurement of 0.5 N

Conclusion

- Prototype can function and trap model lanternflies
 - Experimentation could likely find a configuration that effectively contains living SFL
- Pieces work individually and as a whole:
 - Funnel and flaps can open when 0.5 N are applied, and close to block an escaping model bug
 - Drawer supports expected load of 5.4kg and opens smoothly in ideal conditions
 - Trap can be mounted anywhere from 0 to 9 feet in the air with reasonable stability
- Further testing could investigate:
 - How real SLFs interact with the trap and funnel
 - How the trap handles the vibrations from a harvester
 - How well it attracts SLFs with Tree of Heaven sap in the drawer
- Next iterations should improve upon:
 - The ease of adjusting the pole and placing it in the ground at a large depth for sufficient stability
 - The adjustability of the funnel and flaps
 - Allowing a user to see into the trap from a distance by being transparent
- Design remains low-cost and suitable for field deployment and scaling

Acknowledgements

¹Hayes, Caitlin. "Spotted Lanternflies Could Cost NYS Grape Industry Millions | Cornell Chronicle." Cornell Chronicle, Cornell Chronicle, 24 Jan. 2025, <https://www.cornell.edu/stories/2025/01/spotted-lanternflies-could-cost-nys-grape-industry-millions>.

²Mulhollem. "Grapevines may only need help to survive heavy spotted lanternfly infestations." Penn State University, 2022, <https://www.psu.edu/news/research/story/grapevines-may-only-need-help-survive-heavy-spotted-lanternfly-infestations>. Accessed 2026.

³Pinto, Allan F, et al. "Assessing the Potential Economic Impacts of Spotted Lanternfly (Hemiptera: Fulgoroidea) Infestations on Grape Production in New York State." Journal of Integrated Pest Management, vol. 16, no. 1, 1 Jan. 2025, academic.oup.com/jipm/article/16/1/2/7964417, <https://doi.org/10.1093/jipm/pnae039>.

